TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Chapter 1

CMS – Countermeasures Management Switch

1.1 Concept

The Countermeasures Management Switch (CMS) is a four-direction hat switch mounted on the side-stick controller. It controls the aircraft's defensive avionics: the ALE-47 CMDS (Countermeasures Dispenser Set — the chaff and flare dispensing system) and ECM (Electronic Countermeasures), either external pods or internal avionics, depending on block and variant.

CMS actuation procedures are detailed in Section 1.2. Block and variant differences are covered in Section 1.3.

For CMDS and ECM operation, see *TO 1F-16CMAM-34-1-1*, Dash-34, sections `2.7.1 and 2.7.2`, respectively. Diagrams of the side-stick controller and throttle are provided in Dash-34 § 2.1.5. Figure 1.1 shows the CMS location on the F-16 side-stick controller.

`F-16_Side_Stick_Controller-1.jpg`

Figure 1.1: F-16 Throttle and Side-Stick Controller. Image by Falconpedia (`falcon4.wikidot.com`), via Wikimedia Commons (`https://commons.wikimedia.org/wiki/File:F-16_Side_Stick_Controller.jpg`), licensed under the Creative Commons Attribution-Share Alike 3.0 Unported (CC BY-SA 3.0) license.

1.2 CMS Actuation

CMS actuation is presented here in tables organized by system: CMDS (Manual, Automatic, and Semi-Automatic modes) and ECM (external pods ALQ-131/ALQ-184 and internal IDIAS).

CMS actuation is *independent* of the Master Mode currently selected.

Each table entry specifies:

- Mode: The operational context (Master Mode).
- Direction: CMS hat direction (Up, Down, Left, Right).
- Action: Press type (Short, Long, Long Hold).
- Function: What the CMS command activates or controls.
- Effect / Nuance: Resulting system behavior, including tactics and constraints.
- Dash34: Reference section in the Dash-34 manual.
- Training: Recommended BMS training mission(s) (see BMS Training Manual 4.38.1 for descriptions).

1.2.1 CMDS

The ALE-47 CMDS operates in three modes selected via the CMDS CCU MODE knob: Manual (MAN), Automatic (AUTO), and Semi-Automatic (SEMI). The program executed in each mode — Programs 1–4 — is selected via the PRGM knob on the CMDS panel.

Programs 5 and 6 are independent of the PRGM knob position. Program 5 is activated via the slap switch (on the left side panel, near the throttle), independent of CMS, and is not covered in this guide.

1.2.1.1 Manual Mode

In Manual (MAN) mode, the pilot runs countermeasure programs via **CMS Up** and **CMS Left**.

Table 1.1: CMS Actuation with CMDS Manual Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS MAN	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once per press. No threat sensing; no automatic triggering. Overrides any AUTO dispensing, if running.	2.7.2.2	18, 28
CMDS MAN	Left	Short	Execute Program 6	Program 6 can be dispensed independently of the PRGM knob position.	2.7.2.2	19

1.2.1.2 Automatic Mode

In Automatic (AUTO) mode, the ALE-47 CMDS dispenses the selected program (1–4) *continuously* in response to threats detected by the Radar Warning Receiver (RWR).

Pilot consent is given *once* by pressing **CMS Down**; dispensing continues until **CMS Right** is pressed or expendables are exhausted.

Table 1.2: CMS Actuation with CMDS Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, interrupting any ongoing AUTO dispensing. After execution, AUTO resumes if the threat persists.	2.7.2.2	18, 28
CMDS AUTO	Left	Short	Execute Program 6	Program 6 can be dispensed independently of the PRGM knob position. Overrides AUTO; after execution, AUTO resumes.	2.7.2.2	19
CMDS AUTO	Down	Short	Give Consent; Enable AUTO Dispensing	Grants consent for AUTO dispensing. RWR-detected threats trigger automatic execution of the program selected via the CMDS panel PRGM knob. Dispensing continues until the threat clears or CMS Right is pressed. Consent state persists if the pilot switches to MAN mode; returning to AUTO resumes auto-dispensing if a threat is detected.	2.7.2.1	18, 28
CMDS AUTO	Right	Short	Cancel Consent; Disable AUTO	Removes CMDS consent and places the ALE-47 in Standby. Automatic dispensing halts immediately. Pilot must re-issue CMS Down to resume AUTO operation.	2.7.2.1	18, 28

1.2.1.3 Semi-Automatic Mode

In Semi-Automatic (SEMI) mode, the ALE-47 CMDS dispenses *one program* per RWR-detected threat, with pilot consent required for *each event*.

When a threat is detected, the CMDS control unit displays `DISPENSE RDY` and the system sounds a `COUNTER` voice message.

Table 1.3: CMS Actuation with CMDS Semi-Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS SEMI	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, independent of RWR threat state. After execution, CMDS resumes SEMI monitoring.	2.7.2.2	18, 28
CMDS SEMI	Left	Short	Execute Program 6	Program 6 can be dispensed independently of the PRGM knob position. After execution, CMDS resumes SEMI monitoring.	2.7.2.2	19
CMDS SEMI	Down	Short	Give Consent; Dispense One Program	Executes one instance of the selected program in response to a `COUNTER` prompt. If the threat persists or a new threat appears, CMDS issues `COUNTER` again. Consent state is tracked; switching to AUTO while consent is active resumes auto-dispensing immediately on the next threat.	2.7.2.1	18, 28

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Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS SEMI	Right	Short	Cancel Consent; Return to Standby	Removes SEMI consent. CMDS halts monitoring and returns to Standby. `COUNTER` messages cease.	2.7.2.1	18, 28

1.2.2 ECM

External ECM pods (ALQ-131, ALQ-184) and internal IDIAS (Improved Defensive Internal Avionic System) provide active jamming across radar frequency bands. **CMS Down** grants consent to external pods. **CMS Left** cycles operational modes for internal IDIAS.

1.2.2.1 External ECM Pod

CMS Down grants consent for ECM pod transmission. The XMIT switch on the ECM control panel sets the transmission mode: XMIT 1 (Avionics Priority, AFT antenna only), XMIT 2 (ECM Priority, both FWD and AFT antennas), or XMIT 3 (Active Jam, continuous transmission). The ECM Enable light on the miscellaneous panel reflects consent state. The ECM system is *inhibited* when the pilot moves the RF switch from NORM, or when the landing gear handle is down.

Table 1.4: CMS Actuation with External ECM Pod

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
ECM Pod	Down	Short	Grant ECM Consent	CMS Down illuminates the ECM Enable light. The pod transmits in the mode set by the XMIT switch. Transmission continues until CMS Right is pressed or the RF switch is moved from `NORM`.	2.7.4.1.1, 2.7.4.2.5	28
ECM Pod	Right	Short	Remove ECM Consent	CMS Right removes consent and places the pod in Standby. The ECM Enable light extinguishes. Transmission does not resume until the pilot re-issues CMS Down.	2.7.4.1.1, 2.7.4.2.5	28

1.2.2.2 Internal ECM (IDIAS)

IDIAS provides three operational modes controlled via **CMS Left**: Standby (`STBY`), Avionics Priority (`AVNC`), and ECM Priority (`ECM`). Operational modes are available only when the XMTR button is set to OPER. The active mode is reflected on the HUD, FCR, RWR, and TFR pages.

Table 1.5: CMS Actuation with Internal ECM (IDIAS)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
IDIAS ECM	Left	Short	Cycle ECM Operational Mode	Each short press of CMS Left advances the IDIAS mode: `STBY` → `AVNC` (Avionics Priority) → `ECM` (ECM Priority) → `AVNC` → `ECM`.	2.7.4.1.2	28
IDIAS ECM	Right	Short	Remove ECM Consent	CMS Right sets IDIAS to `STBY`, removing consent. `AVNC` or `ECM` modes are not restored until CMS Left is re-applied.	2.7.4.1.2	28

1.2.3 Consent and Constraints

When equipped with an external ECM pod, **CMS Down** grants consent to both CMDS (in AUTO or SEMI mode) and the external ECM pod *simultaneously*.

Table 1.6: CMS Interaction with CMDS and ECM (Consent and Constraints)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO SEMI + ECM Pod	Down	Short	Joint Consent (CMDS + ECM)	A *single* CMS Down press grants consent to *both* the ALE-47 CMDS (AUTO or SEMI mode) and the external ECM pod.	2.7.2.1, 2.7.4.2.5	18, 28

1.2.4 Operational Notes

State Tracking: In AUTO and SEMI modes, the CMDS retains consent state when the pilot switches to MAN mode. If the pilot re-engages AUTO without pressing **CMS Down**, the CMDS begins dispensing immediately on the next detected threat.

Bingo Quantity Behavior: If expendables fall to or below the bingo quantity, the CMDS requests consent and continues dispensing. The VMU issues a `LOW` message when an expendable reaches the bingo quantity, and an `OUT` message when an expendable is depleted. Dispensing does not stop automatically. The pilot can press **CMS Right** to inhibit AUTO dispensing, or switch to MAN mode to control expenditure directly.

RF Switch Override: When ECM consent is active, moving the RF switch from NORM to QUIET or SILENT places the ECM system in Standby. Returning the RF switch to NORM does *not* restore consent. The pilot must re-issue **CMS Down** to resume ECM transmission.

IDIAS vs. External ECM Pod: The joint consent described in Table 1.6 applies to external ECM pod configurations *only*. On IDIAS-equipped aircraft, **CMS Left** is used to cycle ECM modes; **CMS Down** has no ECM function on these variants.

Ground Safety: On the ground, ECM consent state is Standby. If the pilot holds **CMS Down** on the ground, the ECM pod radiates until the pilot releases **CMS Down**. Ground personnel must be clear of the radiation area if, for any reason, ECM must be enabled on the ground.

1.3 ECM Configuration by Block and Variant

CMS interaction with CMDS is *uniform* across all F-16 blocks and variants (see Section 1.2.1). External Pod and IDIAS F-16 variants, however, use *different* CMS procedures and ECM control panels (see Section 1.2.2):

Table 1.7: ECM Configuration by Variant — CMS Interface Summary

ECM Configuration	CMS Direction	Panel Control
External Pod (ALQ-131/ALQ-184)	CMS Down (consent)	XMIT knob (positions: 1, 2, 3)
Internal IDIAS	CMS Left (mode cycling)	XMTR switch (positions: STBY, OPER)

1.3.1 ECM Configurations in BMS

Falcon BMS 4.38.1 implements two distinct ECM system architectures, each with different CMS actuation methods. Detailed actuation procedures are provided in Section 1.2.

The variants listed in the following tables reflect Dash-34 documentation; not all may be available as selectable aircraft in Falcon BMS 4.38.1.

1.3.1.1 ECM Pods (ALQ-131 / ALQ-184)

These variants use the ECM Pod Control Panel with manual jamming band selection. **CMS Down** grants consent for ECM transmission. For detailed procedures, see Section 1.2.2.1.

Table 1.8: External ECM Pod Variants

Operator	Block/Variant
USAF	Blocks 40/42/50/52 (CM designation)
NATO	Block 15 operators (Belgium, Denmark, Netherlands, Norway)
International	Egypt, Korea KF-16C Block 32

1.3.1.2 Internal ECM (IDIAS)

These variants use the IDIAS ECM control panel. **CMS Left** cycles the operational mode. For detailed procedures, see Section 1.2.2.2.

Table 1.9: Internal ECM (IDIAS) Variants

Operator	Block/Variant
Israel	F-16I Sufa Block 52, Barak I Block 30, Barak II Block 40
International	Greek HAF (Blocks 50 PXII, 52 PXIII, 52+ PXIV Advanced), Korea KF-16C Block 52, Singapore F-16D Block 52